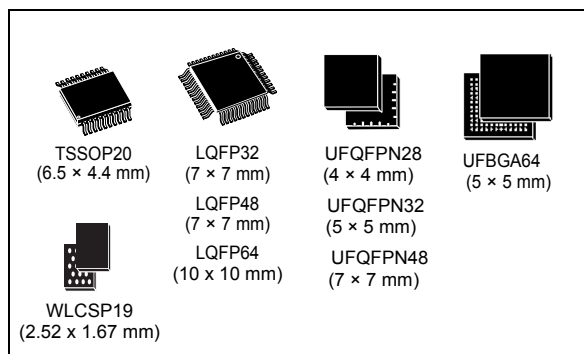


Arm® Cortex®-M0+ 32-bit MCU, 128 KB flash, 24 KB RAM,
2 x USART, timers, ADC, USB FS, comm. I/Fs, 2-3.6V

Datasheet - production data

Features

- Includes ST state-of-the-art patented technology
- Core: Arm® 32-bit Cortex®-M0+ CPU, frequency up to 48 MHz
- 40°C to 85°C/105°C/125°C operating temperature
- Memories
 - Up to 128 Kbytes of flash memory with protection and securable area
 - 24 Kbytes of SRAM with hardware parity check
- CRC calculation unit
- Reset and power management
 - Voltage range: 2.0 V to 3.6 V
 - Separate I/O supply pin (1.65 V to 3.6 V)
 - Power-on / power-down reset (POR/PDR)
 - Programmable brownout reset (BOR)
 - Low-power modes: Sleep, Stop, Standby, Shutdown
- Clock management
 - 4 to 48 MHz crystal oscillator
 - 32 kHz crystal oscillator with calibration
 - Internal 48 MHz RC oscillator (±1 %)
 - Internal 48 MHz RC oscillator with clock recovery system
 - Internal 32 kHz RC oscillator (±5 %)
- Up to 61 fast I/Os
 - All mappable on external interrupt vectors
 - All 5 V-tolerant
- 5-channel DMA controller with flexible mapping
- 12-bit, 0.4 µs ADC (up to 19 ext. channels)
 - Conversion range: 0 to 3.6 V
- 9 timers: 16-bit for advanced motor control, one 32-bit timer and four 16-bit general-purpose, two watchdogs, SysTick timer
- Calendar RTC with alarm



- Communication interfaces
 - Two I²C-bus interface supporting Fast-mode Plus (1 Mbit/s) with extra current sink; one supporting SMBus/PMBus™ and wake-up from Stop mode
 - Two USARTs with master/slave synchronous SPI; one supporting ISO7816 interface, LIN, IrDA capability, auto baud rate detection and wake-up feature
 - Two SPIs (24 Mbit/s) with 4- to 16-bit programmable bitframe, one multiplexed with I²S interface; two extra SPIs through USARTs
- USB 2.0 FS device (crystal-less) and host controller
- Development support: serial wire debug (SWD)
- 96-bit unique ID
- All packages ECOPACK 2 compliant

Table 1. Device summary

Reference	Part number
STM32C071x8	STM32C071F8, STM32C071G8, STM32C071K8, STM32C071C8, STM32C071R8
STM32C071xB	STM32C071FB, STM32C071GB, STM32C071KB, STM32C071CB, STM32C071RB